



Andrey Aguirre Obregón

Senior Software Engineer · .NET · Angular · React · SQL Server

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Heredia, Costa Rica · Remote / hybrid

PROFILE

Senior Software Engineer with 9+ years across banking, healthcare, retail and SaaS. I mostly work with .NET 8, Angular 19, React and SQL Server, using Clean Architecture and CI/CD.

Today I build AI agents on Azure AI Foundry / Google Cloud that serve patients and automate clinic operations, and I ship my own products to production: AI marketing agents, rental management and vehicle history. I have been part of technical teams delivering a bank vehicle-tax (marchamo) payment system that doubled projected sales, while keeping a solid hand on critical integrations like e-invoicing. I round it all out with Machine Learning studies.

EXPERIENCE

Senior Software Engineer — Grupo Malengo

Mar 2024 — Present

- Lead development of AlturaRH, SikuApps and the Admin module: internal HR and clinical-management products.
- Omnichannel CRM and Inbox powered by AI agents on Azure AI Foundry: patient conversations handled, triaged and resolved autonomously, 24/7.
- Automation engine that gives doctors their time back: from simple reminders and tasks to multi-step clinical and administrative workflows, hands-free.
- WhatsApp broadcast system built into the Inbox and CRM: segmented campaigns, centralized replies and full per-patient traceability.
- Internal Customer Success platform for client management and continuity: account health, early risk alerts and end-to-end lifecycle tracking.
- HR and payroll modules, e-invoicing middleware (XML 4.4) and REST integrations with SikuMed (~40% less manual processing) with .NET 8, Angular 19 and SQL Server.

Senior Software Engineer — 3Pillar Global

Feb 2020 — Mar 2024

- Vehicle-tax (marchamo) payment system for Banco Promerica: doubled projected sales volume in its first year.
- Native Android inventory control for a retail app in Canada.
- Credit-line optimization for a high-volume financial institution.
- Distributed teams across the US, Canada and LATAM using Scrum.

Software Engineer — SIKU

Jan 2019 — Feb 2020

- Medical-records software for healthcare professionals nationwide.
- Trello-style board for medical teams with C#, Web API, SQL Server and Angular.
- FoodNotes (web + Android nutrition app), from architecture to production.

Software Engineer — Central Bank of Costa Rica (via Crux Consultores)

Apr 2018 — Jan 2019

- Financial supervision and compliance tools with .NET Framework and SQL Server.
- Technical coordination between compliance, reporting and development to meet regulatory deadlines.

Tech Lead — Exdesa

Jan 2017 — Apr 2018

- E-invoicing platform in Angular + Ionic, compliant with Costa Rican regulations.
- Real-time routing, ordering and cash-closing systems for field sales teams.

PROJECTS

- **Rentify** — Rental management for property owners in Costa Rica: AI-generated contracts, financial dashboard and multi-tenant architecture. (rentifycr.com)
- **Nexora Agents** — An autonomous marketing department: AI agents that research, create and publish social media content; the client only approves. (nexoragents.com)
- **HEC Box** — Website for HEC Box, a Miami-to-Costa Rica package forwarding service: real-time tracking, consolidated shipments from multiple stores and home delivery. (hecboxcr.com)
- **Kardex ID** — Mobile app that predicts mechanical failures with AI, certifies vehicle history with PDF reports, and tracks fuel and maintenance expenses; includes digital ownership transfer and fleet plans. (kardexapp.com/es)
- **DuoPay** — Shared expenses, budgets and balances for couples, on web and mobile.
- **Andara Dashboard** — My own dashboard to run all my products: projects, stages, tasks and production readiness.
- **Algorithmic trading system** — My own Smart Money Concepts engine in Python: detects order blocks, fair value gaps, liquidity sweeps and structure breaks with no look-ahead bias. An ML model manages exits, with fixed per-trade risk, circuit breakers and Telegram alerts; trades crypto futures 24/7 from Docker in the cloud.
- **Roman Agent** — My autonomous personal agent, built on OpenClaw (open source): runs 24/7 in the cloud, understands natural-language instructions and automates everything from everyday tasks to multi-step workflows. Tailored to the way I work and deployed on Railway.
- **Sparrow** — Anime fashion e-commerce: catalog, cart and order management.
- **Club de Leones CR** — Mobile app for the Lions Club of Costa Rica: members, aid requests and activities.
- **Exchange Chronicle** — Trading journal: trade log with a metrics and performance dashboard.

TECH STACK

Backend	.NET 8 · C# · Web API · EF Core · VB.NET · Node.js · Python
Frontend	Angular 19 · React · Next.js · TypeScript · Tailwind CSS · HTML5
Mobile	React Native / Expo · Native Android · Ionic · Flutter
Data	SQL Server · PostgreSQL · MySQL · Supabase
DevOps	Azure DevOps · CI/CD · Git · Docker · IIS · Railway · Vercel
Architecture	Clean Architecture · REST APIs · JWT / OAuth2 · Multi-tenant
AI	AI agents · Gemini API · Azure AI Foundry · AI-assisted development · Machine Learning
Reporting & tax	Reporting Services · Power BI · E-invoicing XML 4.4

EDUCATION

- **Bachelor's in Systems Engineering** — University of Costa Rica (Completed)
 - **Machine Learning Technician** — Cenfotec (In progress · 6 of 8 modules)
 - **AI-assisted programming optimization** — Udemy (Completed)
- Languages:** Spanish (Native) · English (Upper-intermediate (B2))